



Using the associative, commutative and distributive laws together

Task A: Using more than one law

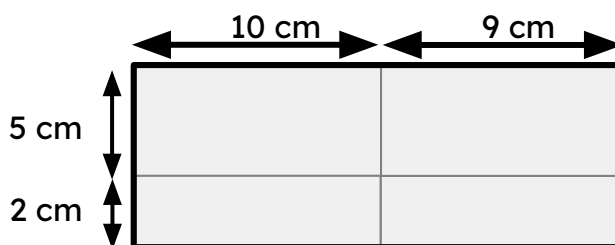
1) Lucas wants to find the area to this rectangle. He starts by writing this calculation:

$$(5 \times 10) + (\square \times 10) + (5 \times \square) + (\square \times \square)$$

a) Complete Lucas' calculation.

$$(5 \times 10) + (2 \times 10) + (5 \times 9) + (2 \times 9)$$

b) Show how Lucas split the rectangle.



c) How many different ways can you split the rectangle to find the same area?

There are so many different ways! Calculations will need to be checked.

2) Fill in the blanks using your knowledge on the distributive law.

a) $7 \times 8 + 12 \times 11$

$$7 \times 4 \times 2 + 3 \times 4 \times 11$$

$$4 \times 14 + 4 \times 33$$

$$4 \times (14 + 33) = 188$$

b) $15 \times 9 + 25 \times 9 + 30 \times 3$

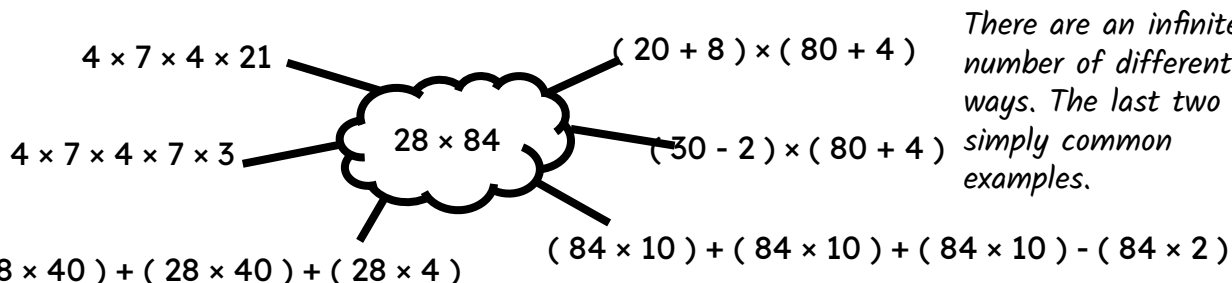
$$15 \times 9 + 25 \times 9 + 10 \times 3 \times 3$$

$$15 \times 9 + 25 \times 9 + 10 \times 9$$

$$9 \times (15 + 25 + 10)$$

$$9 \times (50) = 450$$

3) How many different ways can you split the calculation? Some have been suggested for you to fill in.



There are an infinite number of different ways. The last two are simply common examples.



Task B: Efficiently solve problems

1) Fill in the blanks with your knowledge of the distributive law.

a) $15 \times 3.2 + 10 \times 3.2 - 20 \times 3.2$

$$5 \times 3 \times 3.2 + 5 \times 2 \times 3.2 - 5 \times 4 \times 3.2$$

$$5 \times 9.6 + 5 \times 6.4 - 5 \times 12.8$$

$$5 \times (9.6 + 6.4 - 12.8) = 16$$

2) Are the following statements true or false? Explain how you know.

a) $5 \times 3.2 + 3.2 \times 3 = 2.5 \times 4 \times 3.2$

$$3.2 \times (5 + 3) = 2.5 \times 2 \times 2 \times 3.2$$

False, so they are not equal

$$3.2 \times 8 = 10 \times 3.2$$

$$25.6 \neq 32$$

b) $2 \times (17 \times 3.2 + 1.8 \times 17) = 17^2 - (7 \times 17)$

$$2 \times 17 \times (3.2 + 1.8) = 17 \times 17 - (7 \times 17)$$

True, so they are equal

$$34 \times 5 = 17 \times (17 - 7)$$

$$170 = 170$$

3a) Fill the three gaps in this calculation with the same single-digit number. Try a different single-digit number.

Example

$$(7 \times 5 + 5 \times 3) + 5$$

Choosing 5: $35 + 15 + 5 = 55$

b) What do you notice? Can you explain why this happens?

$$(7 \times \square + \square \times 3) + \square$$

Using the commutative and associative laws we then have

$$(7 \times \square + 3 \times \square + 1 \times \square)$$

Using the distributive law we then have

$$\square \times (7 + 3 + 1)$$

$$\square \times 11$$

We always have the number chosen $\times 11$

